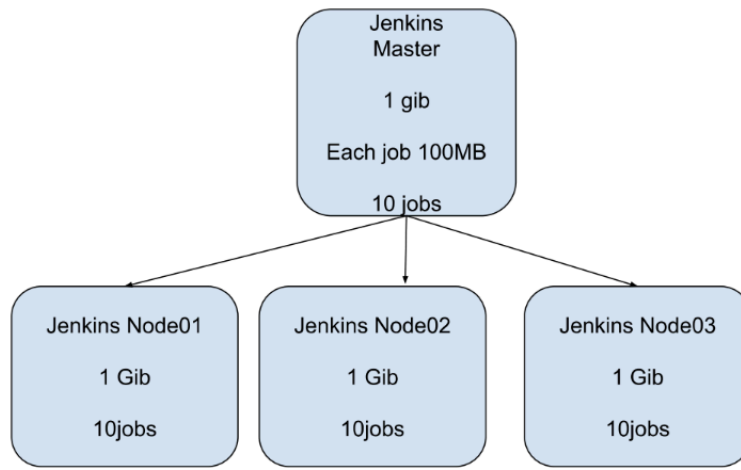




Stage01: install and configure jenkins master

1. Create a Debian 11 "Bullseye" - x64 Gen2 linux Vm
2. `$ sudo -i`
3. `# apt-get update`
4. `# sudo apt-get install openjdk-17-jdk`
5. `# apt-get install maven`
6. `# apt-get install git`
7. `# install jenkins(https://get.jenkins.io/debian-stable/)`
8. add jenkins password
`# passwd jenkins`
9. make jenkins user as sudoer
`# visudo`
`jenkins ALL=(ALL) NOPASSWD: ALL`
Ctr+x, press y and press enter
10. password authentication
`# vi /etc/ssh/sshd_config`
Password authentication yes
:wq
11. restart ssh service
`# service ssh restart`

Stage02: install and configure jenkins nodes

12. Create a Debian 11 "Bullseye" - x64 Gen2 linux Vm
13. \$ sudo -i
14. # apt-get update
15. # sudo apt-get install openjdk-17-jdk
16. # apt-get install maven
17. # apt-get install git
18. # **don't install jenkins**(<https://get.jenkins.io/debian-stable/>)
19. add jenkins password
#adduser jenkins
20. make jenkins user as sudoer
visudo
jenkins ALL=(ALL) NOPASSWD: ALL
Ctrl+x, press y and press enter
21. password authentication
vi /etc/ssh/sshd_config
Password authentication yes
:wq
22. restart ssh service
service ssh restart

Stage03: configure jenkins master with nodes

On jenkins master

23. # su jenkins
24. \$ cd
25. \$ pwd
26. \$ ssh <private ip of nodes>
27. Configure jenkins master with node
28. Create and execute jenkins job on node side.

jforg:

Jenkins Master-Agent Setup Guide

1. Architecture Overview

- **Jenkins Master:**

- Coordinates builds.
- Schedules jobs and assigns them to agents.
- Handles the build pipeline and stores build results.
- Minimum recommended: 4 GB RAM (you had 1 GB, which caused job execution memory issues).
- **Jenkins Agent (Node):**
 - Executes the build steps.
 - Can be distributed (scale horizontally).
 - **Example Setup:**

Master (4GB RAM)

|

|--- Agent Node 1 (1GB RAM)

|--- Agent Node 2 (1GB RAM)

|--- Agent Node 3 (1GB RAM)

2. EC2 Preparation (Ubuntu-based)

- Launch EC2 instances:
 - Master: t2.medium or higher (4 GB RAM minimum).
 - Agents: t2.micro/t2.small depending on load.
- Update and install tools:

```
#!/bin/bash
```

```
echo "🔄 Updating package list..."
```

```
sudo apt update -y
```

echo "📦 Installing OpenJDK 17..."

sudo apt install openjdk-17-jdk -y

echo "📦 Installing OpenJDK 17..."

sudo apt install openjdk-8-jdk

echo "📦 Installing Maven (optional)..."

sudo apt install maven -y

echo "📦 Installing 'tree' utility (optional)..."

sudo apt install tree -y

echo "🔑 Adding Jenkins repository key..."

sudo mkdir -p /usr/share/keyrings

curl -fsSL https://pkg.jenkins.io/debian-stable/jenkins.io-2023.key | sudo tee
/usr/share/keyrings/jenkins-keyring.asc > /dev/null

echo "📦 Adding Jenkins source list..."

echo "deb [signed-by=/usr/share/keyrings/jenkins-keyring.asc]
https://pkg.jenkins.io/debian-stable binary/" | sudo tee /etc/apt/sources.list.d/jenkins.list >
/dev/null

echo "🔄 Updating package list again..."

sudo apt update -y

```
echo "📦 Installing Jenkins..."
```

```
sudo apt install jenkins -y
```

```
echo "🚀 Starting and enabling Jenkins service..."
```

```
sudo systemctl start jenkins
```

```
sudo systemctl enable jenkins
```

```
echo "🛡️ Configuring UFW firewall rules..."
```

```
sudo ufw allow 8080
```

```
sudo ufw allow OpenSSH
```

```
sudo ufw --force enable
```

```
echo "✅ Jenkins installation complete!"
```

```
echo "🌐 Access Jenkins at: http://<your-server-ip>:8080"
```

```
echo "🔑 Initial Admin Password:"
```

```
sudo cat /var/lib/jenkins/secrets/initialAdminPassword
```

```
✅ Jenkins installation complete!
```

```
🌐 Access Jenkins at: http://<your-server-ip>:8080
```

```
🔑 Initial Admin Password:
```

```
cb8549017e714bb9be9beb8a0ca01c6e
```

4. User and SSH Configuration

- Add user `jenkins` and set password:

`sudo adduser jenkins`

`sudo passwd jenkins`
- Edit `/etc/ssh/sshd_config`:

`PasswordAuthentication yes`
- Then restart SSH:

`sudo service ssh restart`
- Setup **passwordless SSH** from master to nodes:

`su - jenkins`

`ssh-keygen`

`ssh-copy-id jenkins@<agent_private_ip>`
- Configure **sudo access** (`visudo`):

`jenkins ALL=(ALL:ALL) NOPASSWD:ALL`

5. Connect Nodes from Jenkins UI

- Jenkins Dashboard → Manage Jenkins → Manage Nodes and Clouds → New Node
 - Name: `node-1`
 - Remote root directory: `/home/jenkins`
 - Usage: Use this node as much as possible
 - Launch method: Launch agents via SSH
 - Host: Private IP of agent
 - Credentials: Jenkins user and password

6. Labeling and Job Binding

- Assign labels like `myjenkinsnode01` to each node.
- In job → Configure → Restrict where this project can be run → Label expression.

7. Verify Job Execution

- Execute `env` or `pwd` in Jenkins jobs to verify whether jobs are running on the node or master.
- Ensure that job workspaces are being created in `/home/jenkins/workspace`.

Best Practices

- Avoid executing heavy jobs on the master. Reserve master for coordination.
- Use dedicated labels for job targeting and scaling.
- Monitor memory/CPU of each node using `htop` or Jenkins monitoring plugins.
- Backup Jenkins config and secrets regularly.

Based on your output from `/etc/passwd`, here's a breakdown of the **users on your Jenkins server**, with focus on which ones are **system vs. human accounts**, and their purpose:

✓ Human/Service Users Summary

Username	UID	Purpose	Login Shell
----------	-----	---------	-------------

root	0	Superuser	/bin/bash
jenkinsuser	1000	Likely manually created user for admin or Jenkins testing	/bin/bash
jenkins	114	Jenkins service account (used by Jenkins process)	/bin/bash

System/Service Users (Non-Login)

These are used by the system or background services (e.g., `_chrony`, `lxd`, etc.) and **cannot log in**.

Username	Shell	Role / Service
<code>_chrony</code>	<code>/usr/sbin/nologin</code>	Chrony NTP service
<code>lxd</code>	<code>/bin/false</code>	LXD container management daemon
<code>sshd</code>	<code>/usr/sbin/nologin</code>	OpenSSH server
<code>systemd-*</code>	<code>/usr/sbin/nologin</code>	System services (network, resolve, etc.)

Many others `/usr/sbin/nologin` or
`/bin/false`

For secure background operation

Key Observations

-  **jenkins**: Jenkins creates this user by default. It owns the Jenkins home directory (`/var/lib/jenkins`) and runs the Jenkins service.
-  **jenkinsuser**: You or someone created this manually. This is a real login user (`/home/jenkinsuser`) and can be used for SSH or manual administration.
-  **System accounts** like `_chrony`, `sshd`, and `lxd` should not be deleted — they support critical services.

Tips

- To check if a user has shell access (i.e., can login):

```
grep '/bin/bash' /etc/passwd
```

- To check current user:

```
whoami
```

- To check who's logged in:

```
who
```

Create a virtual machine ...

✔ Validation passed

 Help me create a low cost VM

Help me create a VM optimized for high availability

Help me choose the right VM size for my workload

Basics	
Subscription	Azure for Students
Resource group	RG-Jenkins
Virtual machine name	JenkinsMaster
Region	East US
Availability options	No infrastructure redundancy required
Zone options	Self-selected zone
Security type	Trusted launch virtual machines
Enable secure boot	Yes
Enable vTPM	Yes
Integrity monitoring	No
Image	Ubuntu Server 22.04 LTS - Gen2
VM architecture	x64
Size	Standard D2s v3 (2 vcpus, 8 GiB memory)
Enable Hibernation	No
Authentication type	Password
Username	jenkinsmaster
Public inbound ports	SSH
Azure Spot	No

Disks

Create a virtual machine ...

✓ Validation passed

 Help me create a low cost VM

Help me create a VM optimized for high availability

Help me choose the right VM size for my workload

Disks

OS disk size	Image default
OS disk type	Premium SSD LRS
Use managed disks	Yes
Delete OS disk with VM	Enabled
Ephemeral OS disk	No

Networking

Virtual network	(new) JenkinsMaster-vnet
Subnet	(new) default (10.0.0.0/24)
Public IP	(new) JenkinsMaster-ip
Accelerated networking	On
Place this virtual machine behind an existing load balancing solution?	No
Delete public IP and NIC when VM is deleted	Disabled

Management

Microsoft Defender for Cloud	None
System assigned managed identity	Off
Login with Microsoft Entra ID	Off

Create a virtual machine ...

✓ Validation passed

 Help me create a low cost VM

Help me create a VM optimized for high availability

Help me choose the right VM size for my workload

Management

Microsoft Defender for Cloud	None
System assigned managed identity	Off
Login with Microsoft Entra ID	Off
Auto-shutdown	Off
Enable periodic assessment	Off
Enable hotpatch	Off
Patch orchestration options	Azure-orchestrated patching (preview): patches will be installed by Azure
Reboot setting	Reboot if required

Monitoring

Alerts	Off
Boot diagnostics	On
Enable OS guest diagnostics	Off
Enable application health monitoring	Off

Advanced

Extensions	None
VM applications	None
Cloud init	No

Create a virtual machine ...

 Validation passed

 Help me create a low cost VM

Help me create a VM optimized for high availability

Help me choose the right VM size for my workload

Subscription	Azure for Students
Resource group	RG-Jenkins
Virtual machine name	JenkinsNode03
Region	East US
Availability options	No infrastructure redundancy required
Zone options	Self-selected zone
Security type	Trusted launch virtual machines
Enable secure boot	Yes
Enable vTPM	Yes
Integrity monitoring	No
Image	Ubuntu Server 22.04 LTS - Gen2
VM architecture	x64
Size	Standard D2s v3 (2 vcpus, 8 GiB memory)
Enable Hibernation	No
Authentication type	Password
Username	jenkinsnode03
Public inbound ports	SSH
Azure Spot	No

Create a virtual machine ...

✓ Validation passed

 Help me create a low cost VM

Help me create a VM optimized for high availability

Help me choose the right VM size for my workload

Disks

OS disk size	Image default
OS disk type	Premium SSD LRS
Use managed disks	Yes
Delete OS disk with VM	Enabled
Ephemeral OS disk	No

Networking

Virtual network	(new) JenkinsNode03-vnet
Subnet	(new) default (10.0.0.0/24)
Public IP	(new) JenkinsNode03-ip
Accelerated networking	On
Place this virtual machine behind an existing load balancing solution?	No
Delete public IP and NIC when VM is deleted	Disabled

Management

Microsoft Defender for Cloud	None
System assigned managed identity	Off

< Previous

Next >

Create

Create a virtual machine ...

✓ Validation passed

 Help me create a low cost VM

Help me create a VM optimized for high availability

Help me choose the right VM size for my workload

Management

Microsoft Defender for Cloud	None
System assigned managed identity	Off
Login with Microsoft Entra ID	Off
Auto-shutdown	Off
Enable periodic assessment	Off
Enable hotpatch	Off
Patch orchestration options	Azure-orchestrated patching (preview): patches will be installed by Azure
Reboot setting	Reboot if required

Monitoring

Alerts	Off
Boot diagnostics	On
Enable OS guest diagnostics	Off
Enable application health monitoring	Off

Advanced

Extensions	None
VM applications	None
Cloud init	No

 JenkinsMaster | Connect ☆ ⋮

Virtual machine

 Search





 Refresh

 Troubleshoot ⌵

 More Options ⌵

 Feedback

-  Overview
-  Activity log
-  Access control (IAM)
-  Tags
-  Diagnose and solve problems
-  Resource visualizer

⌵ Connect

-  **Connect**
-  Bastion
- ⌵ Networking

 Network settings

 Load balancing

 Application security groups

 Network manager
- ⌵ Settings

⌵ Availability + scale

⌵ Security

⌵ Backup + disaster recovery

⌵ Operations

⌵ ...
-  Connecting using

Public IP address | **74.235.253.152**


- Admin username : jenkinsmaster

Port ([change](#)) : 22 [Check access](#) ⓘ

Just-in-time policy : Unsupported by plan ⓘ
- Recommended
-  Local machine Azure portal

SSH using Azure CLI

Quickly connect in browser. Supports Microsoft Entra ID authentication. Private key not required.

Public IP address (74.235.253.152)

Select


- Most common
-  Local machine

Native SSH

No additional software needed. Private key required for connection. Best for those with existing SSH tools.

Public IP address (74.235.253.152)

Select


- ⌵ More ways to connect (3)

JenkinsNode03 | Connect

Virtual machine

Search

Refresh Troubleshoot More Options Feedback

- Overview
- Activity log
- Access control (IAM)
- Tags
- Diagnose and solve problems
- Resource visualizer
- Connect

- Connect
- Bastion
- Networking
 - Network settings
 - Load balancing
 - Application security groups
 - Network manager
- Settings
- Availability + scale
- Security
- Backup + disaster recovery
- Operations

Connecting using
Public IP address | 74.235.5.53

Admin username : jenkinsnode03
Port (change) : 22 [Check access](#) ⓘ
Just-in-time policy : Unsupported by plan ⓘ

Recommended

Local machine Azure portal

SSH using Azure CLI

Quickly connect in browser. Supports Microsoft Entra ID authentication. Private key not required.

Public IP address (74.235.5.53)

Select

Most common

Local machine

Native SSH

No additional software needed. Private key required for connection. Best for those with existing SSH tools.

Public IP address (74.235.5.53)

Select

More ways to connect (3)

It seems like you may be new to launching instances in EC2. Take a walkthrough to learn about EC2, how to launch instances and about best practices

Do not show me this message again Take a walkthrough

Launch an instance [Info](#)

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags [Info](#)

Name

JenkinsNode02

[Add additional tags](#)

▼ Application and OS Images (Amazon Machine Image) [Info](#)

An AMI is a template that contains the software configuration (operating system, application server, and applications) required to launch your instance. Search or browse for AMIs if you don't see what you are looking for below

Search our full catalog including 1000s of application and OS images

Recents Quick Start

Amazon Linux

aws

macOS

Mac

Ubuntu

ubuntu

Windows

Microsoft

Red Hat

Red Hat

SUSE Linux

SUSE

Debian

debian

[Browse more AMIs](#)
Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Ubuntu Server 22.04 LTS (HVM), SSD Volume Type

ami-0c3b809fcf2445b6a (64-bit (x86)) / ami-068da067c3c4ef0d3 (64-bit (Arm))

Free tier eligible

Virtualization: hvm ENA enabled: true Root device type: ebs

Description

Ubuntu Server 22.04 LTS (HVM),EBS General Purpose (SSD) Volume Type. Support available from Canonical (<http://www.ubuntu.com/cloud/services>).

Canonical, Ubuntu, 22.04, amd64 jammy image

Architecture	AMI ID	Publish Date	Username	
64-bit (x86)	ami-0c3b809fcf2445b6a	2025-03-05	ubuntu	Verified provider

▼ Instance type [Info](#) | [Get advice](#)

Instance type

t2.micro

Family: t2 1 vCPU 1 GiB Memory Current generation: true

On-Demand Ubuntu Pro base pricing: 0.0134 USD per Hour On-Demand Linux base pricing: 0.0116 USD per Hour

On-Demand SUSE base pricing: 0.0116 USD per Hour On-Demand Windows base pricing: 0.0162 USD per Hour

On-Demand RHEL base pricing: 0.026 USD per Hour

Free tier eligible

☒ All generations

[Compare instance types](#)

▼ Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - *required*

myawskey

 [Create new key pair](#)

▼ Network settings [Info](#)

[Edit](#)

Network | [Info](#)

vpc-09f48782fa2d324ce

Subnet | [Info](#)

No preference (Default subnet in any availability zone)

Auto-assign public IP | [Info](#)

Enable

[Additional charges apply](#) when outside of [free tier allowance](#)

Firewall (security groups) | [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Create security group

☐ Select existing security group

We'll create a new security group called 'launch-wizard-64' with the following rules:

☒ Allow SSH traffic from

Helps you connect to your instance

Anywhere

0.0.0.0/0

☐ Allow HTTPS traffic from the internet

To set up an endpoint, for example when creating a web server

☐ Allow HTTP traffic from the internet

To set up an endpoint, for example when creating a web server

 Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

×

▼ Configure storage [Info](#)

[Advanced](#)

1x GiB  Root volume, Not encrypted

 Free tier eligible customers can get up to 30 GB of EBS General Purpose (SSD) or Magnetic storage

×

[Add new volume](#)

The selected AMI contains instance store volumes, however the instance does not allow any instance store volumes. None of the instance store volumes from the AMI will be accessible from the instance

 Click refresh to view backup information

The tags that you assign determine whether the instance will be backed up by any Data Lifecycle Manager policies.



JenkinsMaster:

```
○ (base) abdulrehamanshaik@MacBookPro Linux % ssh jenkinsmaster@74.235.253.152
The authenticity of host '74.235.253.152 (74.235.253.152)' can't be established.
ED25519 key fingerprint is SHA256:sMwtFU3fi0XSZc1Eb41en3TxaVeTumA+SbL80iBHIT0.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '74.235.253.152' (ED25519) to the list of known hosts.
jenkinsmaster@74.235.253.152's password:
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-1027-azure x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Mon May 12 18:19:59 UTC 2025

System load:  0.09          Processes:      129
Usage of /:   5.3% of 28.89GB Users logged in:  0
Memory usage: 3%          IPv4 address for eth0: 10.0.0.4
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

jenkinsmaster@JenkinsMaster:~$ sudo -i
root@JenkinsMaster:~#
```

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

root@JenkinsMaster:~# vi jenkins.sh
root@JenkinsMaster:~# chmod +x *.sh
root@JenkinsMaster:~# ./k.sh
🔄 Updating package list...
Hit:1 http://azure.archive.ubuntu.com/ubuntu jammy InRelease
Get:2 http://azure.archive.ubuntu.com/ubuntu jammy-updates InRelease [128 kB]
Get:3 http://azure.archive.ubuntu.com/ubuntu jammy-backports InRelease [127 kB]
Get:4 http://azure.archive.ubuntu.com/ubuntu jammy-security InRelease [129 kB]
Get:5 http://azure.archive.ubuntu.com/ubuntu jammy/universe amd64 Packages [14.1 MB]
Get:6 http://azure.archive.ubuntu.com/ubuntu jammy/universe Translation-en [5652 kB]
Get:7 http://azure.archive.ubuntu.com/ubuntu jammy/universe amd64 c-n-f Metadata [286 kB]
Get:8 http://azure.archive.ubuntu.com/ubuntu jammy/multiverse amd64 Packages [217 kB]
Get:9 http://azure.archive.ubuntu.com/ubuntu jammy/multiverse Translation-en [112 kB]
Get:10 http://azure.archive.ubuntu.com/ubuntu jammy/multiverse amd64 c-n-f Metadata [8372 B]
Get:11 http://azure.archive.ubuntu.com/ubuntu jammy-updates/main amd64 Packages [2556 kB]
Get:12 http://azure.archive.ubuntu.com/ubuntu jammy-updates/main Translation-en [415 kB]
Get:13 http://azure.archive.ubuntu.com/ubuntu jammy-updates/restricted amd64 Packages [3485 kB]
Get:14 http://azure.archive.ubuntu.com/ubuntu jammy-updates/restricted Translation-en [620 kB]
Get:15 http://azure.archive.ubuntu.com/ubuntu jammy-updates/universe amd64 Packages [1203 kB]
Get:16 http://azure.archive.ubuntu.com/ubuntu jammy-updates/universe Translation-en [297 kB]
Get:17 http://azure.archive.ubuntu.com/ubuntu jammy-updates/universe amd64 c-n-f Metadata [28.7 kB]
Get:18 http://azure.archive.ubuntu.com/ubuntu jammy-updates/multiverse amd64 Packages [46.5 kB]
Get:19 http://azure.archive.ubuntu.com/ubuntu jammy-updates/multiverse Translation-en [11.8 kB]
Get:20 http://azure.archive.ubuntu.com/ubuntu jammy-updates/multiverse amd64 c-n-f Metadata [592 B]
Get:21 http://azure.archive.ubuntu.com/ubuntu jammy-backports/main amd64 Packages [68.8 kB]
Get:22 http://azure.archive.ubuntu.com/ubuntu jammy-backports/main Translation-en [11.4 kB]
Get:23 http://azure.archive.ubuntu.com/ubuntu jammy-backports/main amd64 c-n-f Metadata [392 B]
Get:24 http://azure.archive.ubuntu.com/ubuntu jammy-backports/restricted amd64 c-n-f Metadata [116 B]
Get:25 http://azure.archive.ubuntu.com/ubuntu jammy-backports/universe amd64 Packages [30.0 kB]
Get:26 http://azure.archive.ubuntu.com/ubuntu jammy-backports/universe Translation-en [16.5 kB]
Get:27 http://azure.archive.ubuntu.com/ubuntu jammy-backports/universe amd64 c-n-f Metadata [672 B]
Get:28 http://azure.archive.ubuntu.com/ubuntu jammy-backports/multiverse amd64 c-n-f Metadata [116 B]
Get:29 http://azure.archive.ubuntu.com/ubuntu jammy-security/main amd64 Packages [2312 kB]
Get:30 http://azure.archive.ubuntu.com/ubuntu jammy-security/main Translation-en [352 kB]
Get:31 http://azure.archive.ubuntu.com/ubuntu jammy-security/restricted amd64 Packages [3366 kB]
Get:32 http://azure.archive.ubuntu.com/ubuntu jammy-security/restricted Translation-en [601 kB]
Get:33 http://azure.archive.ubuntu.com/ubuntu jammy-security/universe amd64 Packages [974 kB]
Get:34 http://azure.archive.ubuntu.com/ubuntu jammy-security/universe Translation-en [210 kB]
Get:35 http://azure.archive.ubuntu.com/ubuntu jammy-security/universe amd64 c-n-f Metadata [21.7 kB]
Get:36 http://azure.archive.ubuntu.com/ubuntu jammy-security/multiverse amd64 Packages [39.6 kB]
Get:37 http://azure.archive.ubuntu.com/ubuntu jammy-security/multiverse Translation-en [8716 B]
Get:38 http://azure.archive.ubuntu.com/ubuntu jammy-security/multiverse amd64 c-n-f Metadata [368 B]
Fetched 37.4 MB in 5s (7276 kB/s)
📦
```

✅ Jenkins installation complete!

🌐 Access Jenkins at: <http://<your-server-ip>:8080>

🔑 Initial Admin Password:

f0ae3cdb952c4ec89f6d6b6b0e97f133

```
GNU nano 6.2 /etc/sudoers.tmp *
#
# This file MUST be edited with the 'visudo' command as root.
#
# Please consider adding local content in /etc/sudoers.d/ instead of
# directly modifying this file.
#
# See the man page for details on how to write a sudoers file.
#
Defaults      env_reset
Defaults      mail_badpass
Defaults      secure_path="/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/snap/bin"
Defaults      use_pty

# This preserves proxy settings from user environments of root
# equivalent users (group sudo)
#Defaults:%sudo env_keep += "http_proxy https_proxy ftp_proxy all_proxy no_proxy"

# This allows running arbitrary commands, but so does ALL, and it means
# different sudoers have their choice of editor respected.
#Defaults:%sudo env_keep += "EDITOR"

# Completely harmless preservation of a user preference.
#Defaults:%sudo env_keep += "GREP_COLOR"

# While you shouldn't normally run git as root, you need to with etckeeper
#Defaults:%sudo env_keep += "GIT_AUTHOR_* GIT_COMMITTER_*"

# Per-user preferences; root won't have sensible values for them.
#Defaults:%sudo env_keep += "EMAIL DEBEMAIL DEBFULLNAME"

# "sudo scp" or "sudo rsync" should be able to use your SSH agent.
#Defaults:%sudo env_keep += "SSH_AGENT_PID SSH_AUTH_SOCK"

# Ditto for GPG agent
#Defaults:%sudo env_keep += "GPG_AGENT_INFO"

# Host alias specification

# User alias specification

# Cmnd alias specification

# User privilege specification
root    ALL=(ALL:ALL) ALL
jenkins ALL=(ALL)    NOPASSWD: ALL[]
# Members of the admin group may gain root privileges
%admin   ALL=(ALL) ALL

# Allow members of group sudo to execute any command
%sudo    ALL=(ALL:ALL) ALL

# See sudoers(5) for more information on "@include" directives:

@includedir /etc/sudoers.d
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
# This is the sshd server system-wide configuration file. See
# sshd_config(5) for more information.

# This sshd was compiled with PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games

# The strategy used for options in the default sshd_config shipped with
# OpenSSH is to specify options with their default value where
# possible, but leave them commented. Uncommented options override the
# default value.

Include /etc/ssh/sshd_config.d/*.conf

#Port 22
#AddressFamily any
#ListenAddress 0.0.0.0
#ListenAddress ::

#HostKey /etc/ssh/ssh_host_rsa_key
#HostKey /etc/ssh/ssh_host_ecdsa_key
#HostKey /etc/ssh/ssh_host_ed25519_key

# Ciphers and keying
#RekeyLimit default none

# Logging
#SyslogFacility AUTH
#LogLevel INFO

# Authentication:

#LoginGraceTime 2m
#PermitRootLogin prohibit-password
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10

#PubkeyAuthentication yes

# Expect .ssh/authorized_keys2 to be disregarded by default in future.
#AuthorizedKeysFile .ssh/authorized_keys .ssh/authorized_keys2

#AuthorizedPrincipalsFile none

#AuthorizedKeysCommand none
#AuthorizedKeysCommandUser nobody

# For this to work you will also need host keys in /etc/ssh/ssh_known_hosts
#HostbasedAuthentication no
# Change to yes if you don't trust ~/.ssh/known_hosts for
# HostbasedAuthentication
#IgnoreUserKnownHosts no
# Don't read the user's ~/.rhosts and ~/.shosts files
#IgnoreRhosts yes

# To disable tunneled clear text passwords, change to no here!
PasswordAuthentication yes
#PermitEmptyPasswords no

# Change to yes to enable challenge-response passwords (beware issues with
# some PAM modules and threads)
KbdInteractiveAuthentication no

# Kerberos options
#KerberosAuthentication no
#KerberosOrLocalPasswd yes
#KerberosTicketCleanup yes
#KerberosGetAFSToken no

# GSSAPI options
#GSSAPIAuthentication no
#GSSAPICleanupCredentials yes
#GSSAPIStrictAcceptorCheck yes
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
root@JenkinsMaster:~# visudo
root@JenkinsMaster:~# passwd jenkins
New password:
Retype new password:
passwd: password updated successfully
root@JenkinsMaster:~# vi /etc/ssh/sshd_config
root@JenkinsMaster:~# service ssh restart
root@JenkinsMaster:~# []
```


JenkinsNode03 | Connect

Virtual machine

Search

Refresh Troubleshoot More Options Feedback

- Overview
- Activity log
- Access control (IAM)
- Tags
- Diagnose and solve problems
- Resource visualizer
- Connect
 - Connect**
 - Bastion
- Networking
 - Network settings
 - Load balancing
 - Application security groups
 - Network manager
- Settings
- Availability + scale
- Security
- Backup + disaster recovery
- Operations

Connecting using
Public IP address | 74.235.5.53

Admin username : jenkinsnode03
Port (change) : 22 [Check access](#)
Just-in-time policy : Unsupported by plan

Recommended

Local machine Azure portal

SSH using Azure CLI

Quickly connect in browser. Supports Microsoft Entra ID authentication. Private key not required.

Public IP address (74.235.5.53)

Select

Most common

Local machine

Native SSH

No additional software needed. Private key required for connection. Best for those with existing SSH tools.

Public IP address (74.235.5.53)

Select

More ways to connect (3)

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
(base) abdulrehamanshaik@MacBookPro Linux % ssh jenkinsnode03@74.235.5.53
The authenticity of host '74.235.5.53 (74.235.5.53)' can't be established.
ED25519 key fingerprint is SHA256:UP05hHua4kAx4wHho/LoHciF8Z6owIZ7QXJzz5Bff7k.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '74.235.5.53' (ED25519) to the list of known hosts.
jenkinsnode03@74.235.5.53's password:
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-1027-azure x86_64)
```

```
* Documentation:  https://help.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:        https://ubuntu.com/pro
```

System information as of Mon May 12 18:30:20 UTC 2025

```
System load:  0.0      Processes:      126
Usage of /:    5.3% of 28.89GB   Users logged in:  0
Memory usage:  3%      IPv4 address for eth0: 10.0.0.4
Swap usage:    0%
```

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See <https://ubuntu.com/esm> or run: `sudo pro status`

The list of available updates is more than a week old.
To check for new updates run: `sudo apt update`

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in `/usr/share/doc/*/copyright`.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

```
jenkinsnode03@JenkinsNode03:~$ sudo -i
root@JenkinsNode03:~# echo "🔄 Updating package list..."
sudo apt update -y
```

```
echo "🔥 Installing OpenJDK 17..."
sudo apt install openjdk-17-jdk -y
```

```
echo "🔥 Installing OpenJDK 17..."
sudo apt install openjdk-8-jdk
```

```
echo "🔥 Installing Maven (optional)..."
sudo apt install maven -y
```

```
echo "🔥 Installing 'tree' utility (optional)..."
sudo apt install tree -y
```

```
GNU nano 6.2 /etc/sudoers.tmp *
#
# This file MUST be edited with the 'visudo' command as root.
#
# Please consider adding local content in /etc/sudoers.d/ instead of
# directly modifying this file.
#
# See the man page for details on how to write a sudoers file.
#
Defaults        env_reset
Defaults        mail_badpass
Defaults        secure_path="/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/snap/bin"
Defaults        use_pty

# This preserves proxy settings from user environments of root
# equivalent users (group sudo)
#Defaults:%sudo env_keep += "http_proxy https_proxy ftp_proxy all_proxy no_proxy"

# This allows running arbitrary commands, but so does ALL, and it means
# different sudoers have their choice of editor respected.
#Defaults:%sudo env_keep += "EDITOR"

# Completely harmless preservation of a user preference.
#Defaults:%sudo env_keep += "GREP_COLOR"

# While you shouldn't normally run git as root, you need to with etckeeper
#Defaults:%sudo env_keep += "GIT_AUTHOR_* GIT_COMMITTER_*"

# Per-user preferences; root won't have sensible values for them.
#Defaults:%sudo env_keep += "EMAIL DEBEMAIL DEBFULLNAME"

# "sudo scp" or "sudo rsync" should be able to use your SSH agent.
#Defaults:%sudo env_keep += "SSH_AGENT_PID SSH_AUTH_SOCK"

# Ditto for GPG agent
#Defaults:%sudo env_keep += "GPG_AGENT_INFO"

# Host alias specification

# User alias specification

# Cmnd alias specification

# User privilege specification
root    ALL=(ALL:ALL) ALL
jenkins ALL=(ALL) NOPASSWD: ALL
# Members of the admin group may gain root privileges
%admin   ALL=(ALL) ALL

# Allow members of group sudo to execute any command
%sudo   ALL=(ALL:ALL) ALL

# See sudoers(5) for more information on "@include" directives:

@includedir /etc/sudoers.d
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
# This is the sshd server system-wide configuration file. See
# sshd_config(5) for more information.

# This sshd was compiled with PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games

# The strategy used for options in the default sshd_config shipped with
# OpenSSH is to specify options with their default value where
# possible, but leave them commented. Uncommented options override the
# default value.

Include /etc/ssh/sshd_config.d/*.conf

#Port 22
#AddressFamily any
#ListenAddress 0.0.0.0
#ListenAddress ::

#HostKey /etc/ssh/ssh_host_rsa_key
#HostKey /etc/ssh/ssh_host_ecdsa_key
#HostKey /etc/ssh/ssh_host_ed25519_key

# Ciphers and keying
#RekeyLimit default none

# Logging
#SyslogFacility AUTH
#LogLevel INFO

# Authentication:

#LoginGraceTime 2m
#PermitRootLogin prohibit-password
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10

#PubkeyAuthentication yes

# Expect .ssh/authorized_keys2 to be disregarded by default in future.
#AuthorizedKeysFile .ssh/authorized_keys .ssh/authorized_keys2

#AuthorizedPrincipalsFile none

#AuthorizedKeysCommand none
#AuthorizedKeysCommandUser nobody

# For this to work you will also need host keys in /etc/ssh/ssh_known_hosts
#HostbasedAuthentication no
# Change to yes if you don't trust ~/.ssh/known_hosts for
# HostbasedAuthentication
#IgnoreUserKnownHosts no
# Don't read the user's ~/.rhosts and ~/.shosts files
#IgnoreRhosts yes

# To disable tunneled clear text passwords, change to no here!
PasswordAuthentication yes
#PermitEmptyPasswords no

# Change to yes to enable challenge-response passwords (beware issues with
# some PAM modules and threads)
KbdInteractiveAuthentication no

# Kerberos options
#KerberosAuthentication no
#KerberosOrLocalPasswd yes
#KerberosTicketCleanup yes
#KerberosGetAFSToken no

# GSSAPI options
#GSSAPIAuthentication no
#GSSAPICleanupCredentials yes
#GSSAPIStrictAccepterCheck yes
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
root@JenkinsNode03:~# adduser jenkins
Adding user `jenkins' ...
Adding new group `jenkins' (1001) ...
Adding new user `jenkins' (1001) with group `jenkins' ...
Creating home directory `/home/jenkins' ...
Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for jenkins
Enter the new value, or press ENTER for the default
  Full Name []:
  Room Number []:
  Work Phone []:
  Home Phone []:
  Other []:

Is the information correct? [Y/n]
root@JenkinsNode03:~# visudo
root@JenkinsNode03:~# vi /etc/ssh/sshd_config
root@JenkinsNode03:~# service ssh restart
root@JenkinsNode03:~#
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
root@JenkinsMaster:~# visudo
root@JenkinsMaster:~# passwd jenkins
New password:
Retype new password:
passwd: password updated successfully
root@JenkinsMaster:~# vi /etc/ssh/sshd_config
root@JenkinsMaster:~# service ssh restart
root@JenkinsMaster:~# su jenkins
jenkins@JenkinsMaster:/root$ cd
jenkins@JenkinsMaster:~$ pwd
/var/lib/jenkins
jenkins@JenkinsMaster:~$ ssh 10.0.0.4
The authenticity of host '10.0.0.4 (10.0.0.4)' can't be established.
ED25519 key fingerprint is SHA256:SMWtFU3fi0XSZclEb41en3TxaVeTumA+5bL80iBHIT0.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.0.0.4' (ED25519) to the list of known hosts.
jenkins@10.0.0.4's password:
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-1027-azure x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Mon May 12 18:39:49 UTC 2025

System load:  0.0               Processes:            144
Usage of /:   9.1% of 28.89GB   Users logged in:     1
Memory usage: 15%              IPv4 address for eth0: 10.0.0.4
Swap usage:   0%

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
   just raised the bar for easy, resilient and secure K8s cluster deployment.

https://ubuntu.com/engage/secure-kubernetes-at-the-edge

Expanded Security Maintenance for Applications is not enabled.

20 updates can be applied immediately.
16 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

2 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

New release '24.04.2 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

jenkins@JenkinsMaster:~$
```

```
jenkins@JenkinsMaster:~$ hostname -i
10.0.0.4
jenkins@JenkinsMaster:~$ ssh 10.0.0.5
The authenticity of host '10.0.0.5 (10.0.0.5)' can't be established.
ED25519 key fingerprint is SHA256:IG2qSDihcsp/Z3hzhjW2PCdoq6j4SJemdkThsnoyQg0.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.0.0.5' (ED25519) to the list of known hosts.
jenkins@10.0.0.5's password:
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-1027-azure x86_64)
```

```
* Documentation:  https://help.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:        https://ubuntu.com/pro
```

System information as of Mon May 12 18:56:25 UTC 2025

System load:	0.05	Processes:	139
Usage of /:	8.5% of 28.89GB	Users logged in:	1
Memory usage:	5%	IPv4 address for eth0:	10.0.0.5
Swap usage:	0%		

Expanded Security Maintenance for Applications is not enabled.

21 updates can be applied immediately.
16 of these updates are standard security updates.
To see these additional updates run: `apt list --upgradable`

2 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at <https://ubuntu.com/esm>

New release '24.04.2 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in `/usr/share/doc/*/copyright`.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

```
jenkins@JenkinsNode01:~$ hostname -i
10.0.0.5
jenkins@JenkinsNode01:~$ pwd
/home/jenkins
jenkins@JenkinsNode01:~$ exit
logout
Connection to 10.0.0.5 closed.
jenkins@JenkinsMaster:~$ ssh 10.0.0.5
jenkins@10.0.0.5's password:
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-1027-azure x86_64)
```

```
* Documentation:  https://help.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:        https://ubuntu.com/pro
```

System information as of Mon May 12 18:56:25 UTC 2025

System load:	0.05	Processes:	139
Usage of /:	8.5% of 28.89GB	Users logged in:	1
Memory usage:	5%	IPv4 address for eth0:	10.0.0.5
Swap usage:	0%		

Expanded Security Maintenance for Applications is not enabled.

21 updates can be applied immediately.
16 of these updates are standard security updates.
To see these additional updates run: `apt list --upgradable`

2 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at <https://ubuntu.com/esm>

New release '24.04.2 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Last login: Mon May 12 18:56:25 2025 from 10.0.0.4
jenkins@JenkinsNode01:~\$

Help me copy this VM in any region

```
jenkins@JenkinsMaster:~$ sudo su - jenkins
jenkins@JenkinsMaster:~$ ls -l ~/.ssh/id_rsa.pub
ls: cannot access '/var/lib/jenkins/.ssh/id_rsa.pub': No such file or directory
jenkins@JenkinsMaster:~$ ssh-keygen -t rsa -b 4096 -C "jenkins@JenkinsMaster"
Generating public/private rsa key pair.
Enter file in which to save the key (/var/lib/jenkins/.ssh/id_rsa):
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /var/lib/jenkins/.ssh/id_rsa
Your public key has been saved in /var/lib/jenkins/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:g0az6Es01V0m7a5GsbWjJ5gQGfmuShcWlH99/w645U jenkins@JenkinsMaster
The key's randomart image is:
+-----[RSA 4096]-----+
|
|..+
|..B o +
|..O =.+
|..O B o+S
|..+ ++o
|..+ +. =. . E
|..+ = 0=0. 00.
|..B..++
+-----[SHA256]-----+
jenkins@JenkinsMaster:~$ cat /var/lib/jenkins/.ssh/id_rsa.pub
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQCAQcRdF0EG6meMohokYlg1uDD548Ue+luNNMR4SUKf1x3oFYEPFKiztc7gl48CoG7wMR1AP1oaJmBXXIyShBa9lwCSTj5XbTTDechYychN0LSjDannRjCoaps7n9qmuh302MaSGY9KUKIUKMj1D9uucogfjVJnFaTDFImqXECTXftE4eCU
x1KwBLu/ySyk/yMyo+YpxAr3glv295ImKJ1Y3Ka+YgY0omK/enLumrkpcIjhs1UCwP20780xHSZcVzF1ggopoiQARFOMdayd0/745xNaacsMaxrfjD28H2MM4bb5lY2x2f4YmubnaVfCBj8vP1s1Segn47Imx75fop18japbD1K+o25Gpu0EB18lpQG1EAmgfbh11G50ddy1I
tHS1x1SjGjntz0F+0lGcG6odh1xnfusevFqM08f7n8072Inkce0867/nNh451laQ0d1jSbAV86AaAnRV0v1QCa1DpqqQV1X2X0J21fjuts1S1wP9hq9RV02rrjycd7BnH5dsstckh0T8Se1R/bA31fnj3T1UN4hKqkQ0l8g/Rvnop/mwyEC149PvrT++vuoG1mw1Skazrade6
RpyopHDeRyp02x0B18U4lMuX01Kj7bXk1SjqlYdew+e3fy27hp0jooBVMUR8bt7Zmn/g9dQ== jenkins@JenkinsMaster
jenkins@JenkinsMaster:~$ []
```

You're trying to SSH from your **JenkinsMaster (Azure)** to **JenkinsNode02 (AWS)** as user **jenkins**, but getting:

Permission denied (publickey)

This error means:

- The **SSH public key** of user **jenkins** on the JenkinsMaster has **not been added** to the **~/.ssh/authorized_keys** file of user **jenkins** on the JenkinsNode02 (AWS).

 Step-by-Step Fix: Setup SSH Access from Azure JenkinsMaster → AWS JenkinsNode

 1. On JenkinsMaster (Azure):

Check for SSH key:

`sudo su - jenkins`

`ls -l ~/.ssh/id_rsa.pub`

If it doesn't exist, generate one:

```
ssh-keygen -t rsa -b 4096 -C "jenkins@JenkinsMaster"
```

Just press enter for defaults (no passphrase).

Copy the public key:

```
cat ~/.ssh/id_rsa.pub
```

🔧 2. On JenkinsNode02 (AWS):

Login with a working user (e.g., `ec2-user`, `ubuntu`, or `root`) using your existing AWS key pair:

```
ssh -i your-aws-key.pem ec2-user@18.188.135.135
```

Then:

Create the `jenkins` user (if not exists):

```
sudo useradd -m -s /bin/bash jenkins
```

-

Create `.ssh` directory:

```
sudo mkdir -p /home/jenkins/.ssh
```

```
sudo nano /home/jenkins/.ssh/authorized_keys
```

-
- Paste the **public key** you copied from Azure JenkinsMaster.

Set permissions:

```
sudo chown -R jenkins:jenkins /home/jenkins/.ssh
```

```
sudo chmod 700 /home/jenkins/.ssh
```



```
sudo chmod 600 /home/jenkins/.ssh/authorized_keys
```

-

✓ 3. Test the SSH from Azure JenkinsMaster

Back on your Azure master:

```
ssh jenkins@18.188.135.135
```

It should log in **without password**.

```
root@ip-172-31-23-64:~# adduser jenkins
Adding user `jenkins' ...
Adding new group `jenkins' (1001) ...
Adding new user `jenkins' (1001) with group `jenkins' ...
Creating home directory `/home/jenkins' ...
Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for jenkins
Enter the new value, or press ENTER for the default
    Full Name []:
    Room Number []:
    Work Phone []:
    Home Phone []:
    Other []:
Is the information correct? [Y/n]
root@ip-172-31-23-64:~# visudo
root@ip-172-31-23-64:~# vi /etc/ssh/sshd_config
root@ip-172-31-23-64:~# service ssh restart
root@ip-172-31-23-64:~# sudo mkdir -p /home/jenkins/.ssh
sudo vi /home/jenkins/.ssh/authorized_keys
root@ip-172-31-23-64:~# cat /home/jenkins/.ssh/authorized_keys
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQACrdF0EG6meWohokYLG1uD0548Ue+luNNWR4SUKf1x3oFYEPFXiztc7gL4BCoG7WMRIAPioajMbXKIy5HaBa9lWCSTj5XbTTDEchYcdN0LSljDa+nRjCoaps7n9qmuH3D2MaSGY9kUKiUkMj1D9uCoqfjVJnFaTDfImqXECTXfTe4eCU
xTkWBLu/ySyk/yMyD+YPxxAr3glv2991mNKJrY3Ka+YgV0omK/0nIuwrKqPcIJh5jUCwP20700xHX52cVzFignGpoIQAr0MDMdayqQ/r4SxNaac9MaXrYjDZ0HzM4b65ljY2x2F4YmubmaVfc6J8vP1sL9egn4f1mx75fopJBjaMpDIK+o2SGpu0EB1R8lpQC61EuMgfhhjiiG50ddyI1
tH3Tx1S1GjHtzQ0F+rdQlgC2GS0EhIxAnFusevFquM08fH8072JnKceYD86T/fNhH6j1aQuDfjsDaV06UahNRV6VvtQCbTDpoqqwViXZKJZlfjutS1S/wP9hq9RVQ2frjycd7BrH50ssNckh0T8Se1R/bAJ1Fnj3TLLNahKGX0QLBg/Rvnop/mwyECi49PVFT++vuoGImxM1SkaZradeG
RpyopHDeRypdM2eXB18U4UuXQ11Kj7bxkiSjqljyde++e3fy27hp+pdJoobVDNiR8bbTZhm/g0dQ= jenkins@JenkinsMaster
root@ip-172-31-23-64:~# sudo chown -R jenkins:jenkins /home/jenkins/.ssh
root@ip-172-31-23-64:~# sudo chmod 700 /home/jenkins/.ssh
root@ip-172-31-23-64:~# sudo chmod 600 /home/jenkins/.ssh/authorized_keys
root@ip-172-31-23-64:~#
```

```
(base) abdulrehamanshaik@MacBookPro Linux % ssh jenkinsmaster@74.235.253.152
jenkinsmaster@74.235.253.152's password:
```

```
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-1027-azure x86_64)
```

```
* Documentation:  https://help.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:        https://ubuntu.com/pro
```

```
System information as of Mon May 12 19:22:25 UTC 2025
```

```
System load:  0.0          Processes:      145
Usage of /:    9.1% of 28.89GB   Users logged in:  2
Memory usage: 15%          IPv4 address for eth0: 10.0.0.4
Swap usage:   0%
```

```
* Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
  just raised the bar for easy, resilient and secure K8s cluster deployment.
```

```
https://ubuntu.com/engage/secure-kubernetes-at-the-edge
```

```
Expanded Security Maintenance for Applications is not enabled.
```

```
20 updates can be applied immediately.
16 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable
```

```
2 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm
```

```
New release '24.04.2 LTS' available.
Run 'do-release-upgrade' to upgrade to it.
```

```
Last login: Mon May 12 18:49:35 2025 from 98.156.1.186
```

```
jenkinsmaster@jenkinsmaster:~$ sudo -i
root@jenkinsmaster:~# su jenkins
jenkins@jenkinsmaster:/root$ cd
jenkins@jenkinsmaster:~$ ssh jenkins@18.188.135.135
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-1024-aws x86_64)
```

```
* Documentation:  https://help.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:        https://ubuntu.com/pro
```

```
System information as of Mon May 12 19:25:36 UTC 2025
```

```
System load:  0.0          Processes:      111
Usage of /:    34.0% of 7.57GB   Users logged in:  2
Memory usage: 26%          IPv4 address for eth0: 172.31.23.64
Swap usage:   0%
```

```
Expanded Security Maintenance for Applications is not enabled.
```

```
70 updates can be applied immediately.
48 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable
```

```
2 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm
```

```
New release '24.04.2 LTS' available.
Run 'do-release-upgrade' to upgrade to it.
```

```
Last login: Mon May 12 19:14:48 2025 from 74.235.253.152
```

```
jenkins@ip-172-31-23-64:~$ hostname -i
172.31.23.64
```

```
jenkins@ip-172-31-23-64:~$
```

(base) abdulrehamanshaik@MacBookPro Linux % ssh jenkinsmaster@74.235.253.152

jenkinsmaster@74.235.253.152's password:

Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-1027-azure x86_64)

* Documentation: <https://help.ubuntu.com>

* Management: <https://landscape.canonical.com>

* Support: <https://ubuntu.com/pro>

System information as of Mon May 12 19:22:25 UTC 2025

System load: 0.0 Processes: 145
Usage of /: 9.1% of 28.89GB Users logged in: 2
Memory usage: 15% IPv4 address for eth0: 10.0.0.4
Swap usage: 0%

* Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s just raised the bar for easy, resilient and secure K8s cluster deployment.

<https://ubuntu.com/engage/secure-kubernetes-at-the-edge>

Expanded Security Maintenance for Applications is not enabled.

20 updates can be applied immediately.

16 of these updates are standard security updates.

To see these additional updates run: `apt list --upgradable`

2 additional security updates can be applied with ESM Apps.

Learn more about enabling ESM Apps service at <https://ubuntu.com/esm>

New release '24.04.2 LTS' available.

Run 'do-release-upgrade' to upgrade to it.

Last login: Mon May 12 18:49:35 2025 from 98.156.1.186

jenkinsmaster@JenkinsMaster:~\$ sudo -i

root@JenkinsMaster:~# su jenkins

jenkins@JenkinsMaster:/root\$ cd

jenkins@JenkinsMaster:~\$ ssh jenkins@18.188.135.135

Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-1024-aws x86_64)

* Documentation: <https://help.ubuntu.com>

* Management: <https://landscape.canonical.com>

* Support: <https://ubuntu.com/pro>

System information as of Mon May 12 19:25:36 UTC 2025

System load: 0.0 Processes: 111

Usage of /: 34.0% of 7.57GB Users logged in: 2

Memory usage: 26% IPv4 address for eth0: 172.31.23.64

Swap usage: 0%

Expanded Security Maintenance for Applications is not enabled.

70 updates can be applied immediately.

48 of these updates are standard security updates.

To see these additional updates run: `apt list --upgradable`

2 additional security updates can be applied with ESM Apps.

Learn more about enabling ESM Apps service at <https://ubuntu.com/esm>

New release '24.04.2 LTS' available.

Run 'do-release-upgrade' to upgrade to it.

Last login: Mon May 12 19:14:48 2025 from 74.235.253.152

jenkins@ip-172-31-23-64:~\$ `hostname -i`

172.31.23.64

jenkins@ip-172-31-23-64:~\$

Home > Compute infrastructure | Virtual machines > JenkinsMaster

JenkinsMaster | Network settings

Virtual machine

Search

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Connect

Connect

Bastion

Networking

Network settings

Load balancing

Application security groups

Network manager

Settings

Availability + scale

Security

Backup + disaster recovery

Operations

This is a new experience. [Please provide feedback](#)

Network security group JenkinsMaster-nsg (attached to networkInterface: jenkinsmaster675)

Impacts 0 subnets, 1 network interfaces

+ Create port rule

Search rules

Source == all Destination == all Protocol == all Action == all

Priority ↑	Name	Port	Protocol	Source	Destination	Action
Inbound port rules (5)						
300	SSH	22	TCP	Any	Any	Allow
310	AllowAnyCustomAnyInbound	Any	Any	Any	Any	Allow
65000	AllowVnetInBound	Any	Any	VirtualNetwork	VirtualNetwork	Allow
65001	AllowAzureLoadBalancerInBound	Any	Any	AzureLoadBalancer	Any	Allow
65500	DenyAllInBound	Any	Any	Any	Any	Deny
Outbound port rules (4)						
320	AllowAnyCustomAnyOutbound	Any	Any	Any	Any	Allow
65000	AllowVnetOutBound	Any	Any	VirtualNetwork	VirtualNetwork	Allow
65001	AllowInternetOutBound	Any	Any	Any	Internet	Allow
65500	DenyAllOutBound	Any	Any	Any	Any	Deny

```
root@JenkinsMaster:~# sudo cat /var/lib/jenkins/secrets/initialAdminPassword
```

```
f0ae3cdb952c4ec89f6d6b6b0e97f133
```

```
root@JenkinsMaster:~#
```

Unlock Jenkins

To ensure Jenkins is securely set up by the administrator, a password has been written to the log ([not sure where to find it?](#)) and this file on the server:

```
/var/lib/jenkins/secrets/initialAdminPassword
```

Please copy the password from either location and paste it below.

Administrator password

Create First Admin User

Username

Admin

Password

.....

Confirm password

.....

Full name

Abdul Rehaman Shaik

E-mail address

abdulrehaman.sk7891@gmail.com

New node

Node name

Node01

Type

☒

 Permanent Agent

Adds a plain, permanent agent to Jenkins. This is called "permanent" because Jenkins doesn't provide higher level of integration with these agents, such as dynamic provisioning. Select this type if no other agent types apply — for example such as when you are adding a physical computer, virtual machines managed outside Jenkins, etc.

Create

Name ?

Node01

Description ?

Plain text Preview

Number of executors ?

1

Remote root directory ?

/home/jenkins

Labels ?

mylabel01

Usage ?

Use this node as much as possible

Launch method ?

Launch agents via SSH

Host ?

10.0.0.5

Credentials ?

- none -

+ Add

🔴

 The selected credentials cannot be found

Host Key Verification Strategy ?

Known hosts file Verification Strategy

Save

Description ?

Plain text

Number

1

Remote

/home/

Labels

mylabel

Usage

Use this

Launch

Launch

Host

10

Cre

-

+

Host

Ke

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Availability ?

Keep this agent online as much as possible

Jenkins Credentials Provider: Jenkins

Add Credentials

Domain

Global credentials (unrestricted)

Kind

Username with password

Scope ?

Global (Jenkins, nodes, items, all child items, etc)

Username ?

jenkins

☐ Treat username as secret ?

Password ?

.....

ID ?

myinstance01

An internal unique ID by which these credentials are identified from jobs and other configuration. Normally left blank, in which case an ID will be generated, which is fine for jobs created using visual forms. Useful to specify explicitly when using credentials from scripted configuration.

(from Credentials Plugin)

Description ?

Azureinstance

Cancel

Add

Description ?

Plain text Preview

Number of executors ?

1

Remote root directory ?

/home/jenkins

Labels ?

mylabel01

Usage ?

Use this node as much as possible

Launch method ?

Launch agents via SSH

Host ?

10.0.0.5

Credentials ?

- current -

+ Add

Host Key Verification Strategy ?

Known hosts file Verification Strategy

Advanced

Availability ?

Keep this agent online as much as possible

Save

🖥️ Nodes

☁️ Clouds

Build Queue

No builds in the queue.

Build Executor Status

🖥️ Built-In Node

0/2

🖥️ Node01

0/1

Nodes

+ New Node

Configure Monitors

🔄

S	Name ↓	Architecture	Clock Difference	Free Disk Space	Free Swap Space	Free Temp Space	Response Time
🖥️	Built-In Node	Linux (amd64)	In sync	26.24 GiB	🔴 0 B	26.24 GiB	0ms ⚙️
🖥️	Node01		N/A	N/A	N/A	N/A	N/A ⚙️
Data obtained		3 ms	4 ms	3 ms	6 ms	3 ms	16 min

Icon:

S

M

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Legend

Configure

- General
- Source Code Management
- Triggers
- Environment
- Build Steps
- Post-build Actions

General

Enabled 

Description

Plain text [Preview](#)

- ☐ Discard old builds [?](#)
- ☐ GitHub project
- ☐ This project is parameterized [?](#)
- ☐ Throttle builds [?](#)
- ☐ Execute concurrent builds if necessary [?](#)
- ☒ Restrict where this project can be run [?](#)

Label Expression [?](#)

mylabel01

[Label mylabel01](#) matches 1 node. Permissions or other restrictions provided by plugins may further reduce that list.

Advanced 

Source Code Management

Connect and manage your code repository to automatically pull the latest code for your builds.

- ☐ None
- ☒ Git [?](#)

Repositories [?](#)

Repository URL [?](#)

https://github.com/AbdulRehman082493/mahalogin.git

Credentials [?](#)

- none -

+

add

Save

Apply

Configure

- General
- Source Code Management
- Triggers
- Environment
- Build Steps
- Post-build Actions

(Auto)

Additional Behaviours

Add

Triggers

Set up automated actions that start your build based on specific events, like code changes or scheduled times.

- ☐ Trigger builds remotely (e.g., from scripts)
- ☐ Build after other projects are built
- ☐ Build periodically
- ☒ GitHub hook trigger for GITScm polling
- ☐ Poll SCM

Environment

Configure settings and variables that define the context in which your build runs, like credentials, paths, and global parameters.

- ☐ Delete workspace before build starts
- ☐ Use secret text(s) or file(s)
- ☒ Add timestamps to the Console Output
- ☐ Inspect build log for published build scans
- ☐ Terminate a build if it's stuck
- ☐ With Ant

Build Steps

Automate your build process with ordered tasks like code compilation, testing, and deployment.

Invoke top-level Maven targets

Goals

clean install

Advanced

Add build step

Post-build Actions

Save

Apply

Timestamps View as plain text

☒ System clock time

☒ Use browser timezone

☐ Elapsed time

☐ None

```
Progress (2): 94/230 kB | 12 kB
Progress (2): 98/230 kB | 12 kB
Progress (2): 102/230 kB | 12 kB
Progress (2): 106/230 kB | 12 kB
Progress (2): 111/230 kB | 12 kB
Progress (2): 115/230 kB | 12 kB
Progress (2): 119/230 kB | 12 kB
Progress (2): 123/230 kB | 12 kB
Progress (2): 127/230 kB | 12 kB

Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-digest/1.0/plexus-digest-1.0.jar (12 kB at 1.1 MB/s)
Progress (1): 131/230 kB
Progress (1): 135/230 kB
Progress (1): 139/230 kB
Progress (1): 143/230 kB
Progress (1): 147/230 kB
Progress (1): 152/230 kB
Progress (1): 156/230 kB
Progress (1): 160/230 kB
Progress (1): 164/230 kB
Progress (1): 168/230 kB
Progress (1): 172/230 kB
Progress (1): 176/230 kB
Progress (1): 180/230 kB
Progress (1): 184/230 kB
Progress (1): 188/230 kB
Progress (1): 193/230 kB
Progress (1): 197/230 kB
Progress (1): 201/230 kB
Progress (1): 205/230 kB
Progress (1): 209/230 kB
Progress (1): 213/230 kB
Progress (1): 217/230 kB
Progress (1): 221/230 kB
Progress (1): 225/230 kB
Progress (1): 229/230 kB
Progress (1): 230 kB

Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-utils/3.0.5/plexus-utils-3.0.5.jar (230 kB at 13 MB/s)
[1;34mINFO [m] Installing /home/jenkins/workspace/JobA/target/mahalogin-3.0.war to /home/jenkins/.m2/repository/com/maha/mahalogin/3.0/mahalogin-3.0.war
[1;34mINFO [m] Installing /home/jenkins/workspace/JobA/pom.xml to /home/jenkins/.m2/repository/com/maha/mahalogin/3.0/mahalogin-3.0.pom
[1;34mINFO [m] [1m----- [m
[1;34mINFO [m] [1;32mBUILD SUCCESS [m
[1;34mINFO [m] [1m----- [m
[1;34mINFO [m] Total time: 10.818 s
[1;34mINFO [m] Finished at: 2025-05-12T19:43:04Z
[1;34mINFO [m] [1m----- [m
Finished: SUCCESS
```


-  Status
-  Delete Agent
-  Configure
-  Build History
-  Load Statistics
-  Script Console
-  Log
-  System Information
-  Disconnect
- Build Executor Status

0/1

Build History on Node01

This history is not guaranteed to include all subtasks executed on the node, e.g. Jenkins Pipeline subtasks will not be displayed.

S	Build	Time Since ↑	Status
✓	JobA #1	1 min 37 sec	stable

Icon: S M L

```
root@JenkinsNode01:~# su jenkins
jenkins@JenkinsNode01:/root$ cd
jenkins@JenkinsNode01:~$ pwd
/home/jenkins
jenkins@JenkinsNode01:~$ cd /home/jenkins
jenkins@JenkinsNode01:~$ ls
remoting  remoting.jar  workspace
jenkins@JenkinsNode01:~$ ls -al
total 1408
drwxr-xr-x 6 jenkins jenkins 4096 May 12 19:42 .
drwxr-xr-x 4 root    root    4096 May 12 18:54 ..
-rw-r--r-- 1 jenkins jenkins  32 May 12 19:00 .bash_history
-rw-r--r-- 1 jenkins jenkins 220 May 12 18:54 .bash_logout
-rw-r--r-- 1 jenkins jenkins 3771 May 12 18:54 .bashrc
drwx----- 3 jenkins jenkins 4096 May 12 19:40 .cache
drwxrwxr-x 3 jenkins jenkins 4096 May 12 19:42 .m2
-rw-r--r-- 1 jenkins jenkins 887 May 12 18:54 .profile
drwxrwxr-x 4 jenkins jenkins 4096 May 12 19:40 remoting
-rw-rw-r-- 1 jenkins jenkins 1396936 May 12 19:40 remoting.jar
drwxrwxr-x 3 jenkins jenkins 4096 May 12 19:42 workspace
jenkins@JenkinsNode01:~$ []
```

Capture

New node

Node name

Type



Permanent Agent

Adds a plain, permanent agent to Jenkins. This is called "permanent" because Jenkins doesn't provide higher level of integration with these agents, such as dynamic provisioning. Select this type if no other agent types apply — for example such as when you are adding a physical computer, virtual machines managed outside Jenkins, etc.



Copy Existing Node

Create

Jenkins

Dashboard

Manage Jenkins

Nodes

Abdul Rehaman Shaik

log out

Name

Node02

Description

Plain text

Number of nodes

1

Remote name

/home/

Labels

mylabel

Usage

Use this node

Launch mode

Launch timeout

Host name

18

Created

+

new

Jenkins Credentials Provider: Jenkins

Add Credentials

Domain

Global credentials (unrestricted)

Kind

Username with password

Scope

Global (Jenkins, nodes, items, all child items, etc)

Username

jenkins

☐ Treat username as secret

Password

.....

ID

myinstance02

Description

AWSInstance

Cancel

Add

The selected credentials cannot be found

Name ?

Node02

Description ?

Plain text

[Preview](#)

Number of executors ?

1

Remote root directory ?

/home/jenkins

Labels ?

mylabel02

Usage ?

Use this node as much as possible

Launch method ?

Launch agents via SSH

Host ?

18.188.135.135

Credentials ?

jenkins/***** (AWSInstance)

+ Add

Host Key Verification Strategy ?

Known hosts file Verification Strategy

Save

Dashboard > Manage Jenkins > Nodes >

Nodes

Clouds

Build Queue

No builds in the queue.

Build Executor Status

Built-In Node

0/2

Node01

0/1

Node02

offline

Nodes

+ New Node

Configure Monitors

S	Name ↓	Architecture	Clock Difference	Free Disk Space	Free Swap Space	Free Temp Space	Response Time
	Built-In Node	Linux (amd64)	In sync	26.07 GiB	10B	26.07 GiB	0ms 
	Node01	Linux (amd64)	In sync	26.25 GiB	10B	26.25 GiB	31ms 
	Node02		N/A	N/A	N/A	N/A	N/A 
Data obtained		14 sec	14 sec	14 sec	14 sec	14 sec	14 sec

Icon:

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Legend



Status

🗑️ Delete Agent

⚙️ Configure

📜 Build History

📈 Load Statistics

📄 Log

Build Executor Status0/1 ▾

🔴 Node02offline

Agent Node02

✎ Add description

Launch agent

Mark this node temporarily offline

ⓘ

🚫 This agent is offline because Jenkins failed to launch the agent process on it. See log for more details

Monitoring Data ▾

Labels

mylabel02

Projects tied to Node02

None

